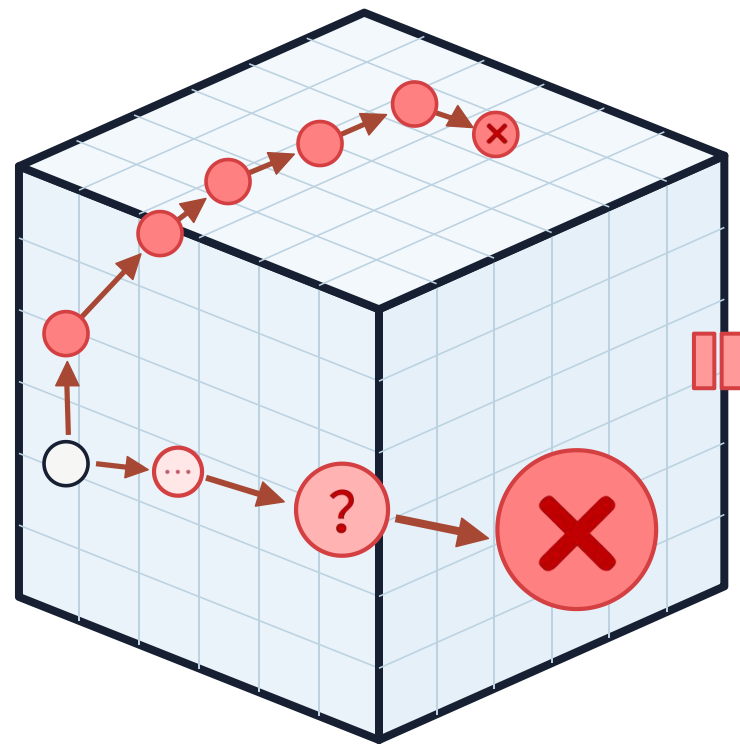


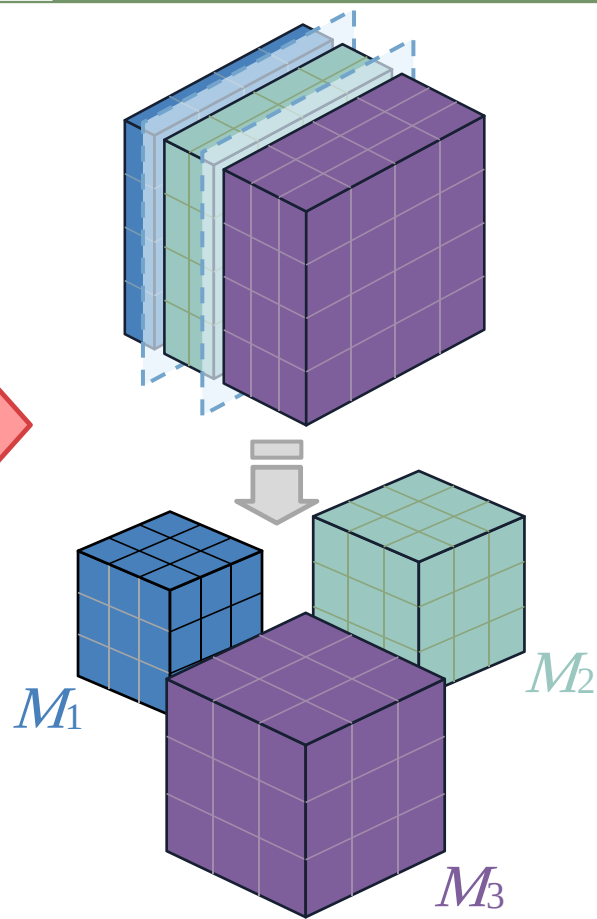
Single-Model Reasoning



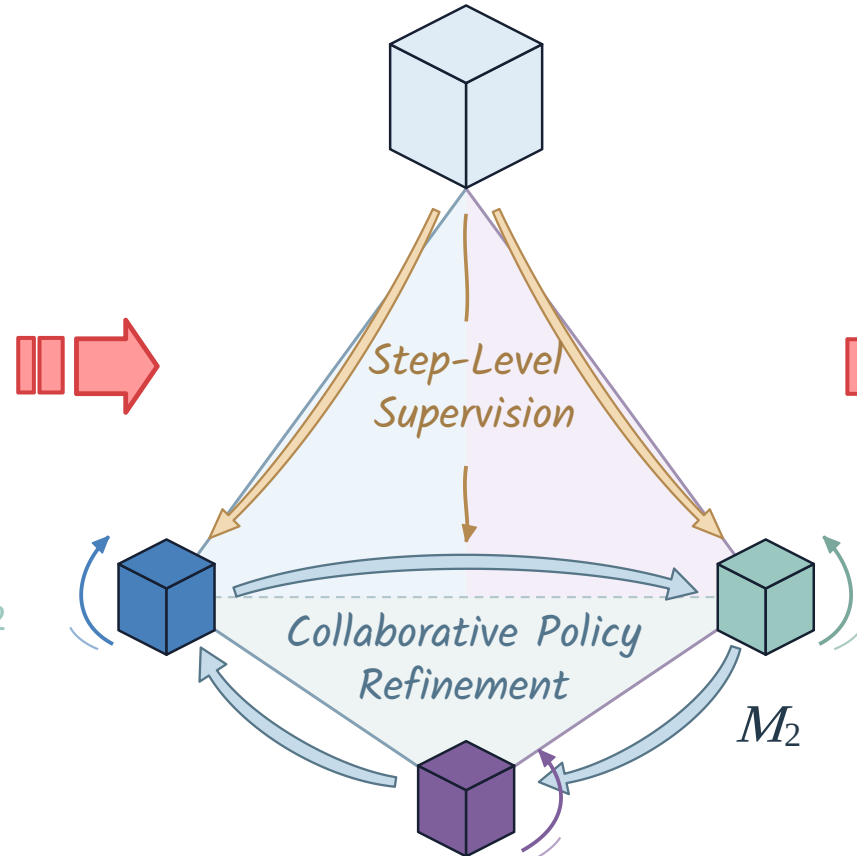
Original Large model M_L

- Limited reassessment
- Error propagation

Collaborative Learning

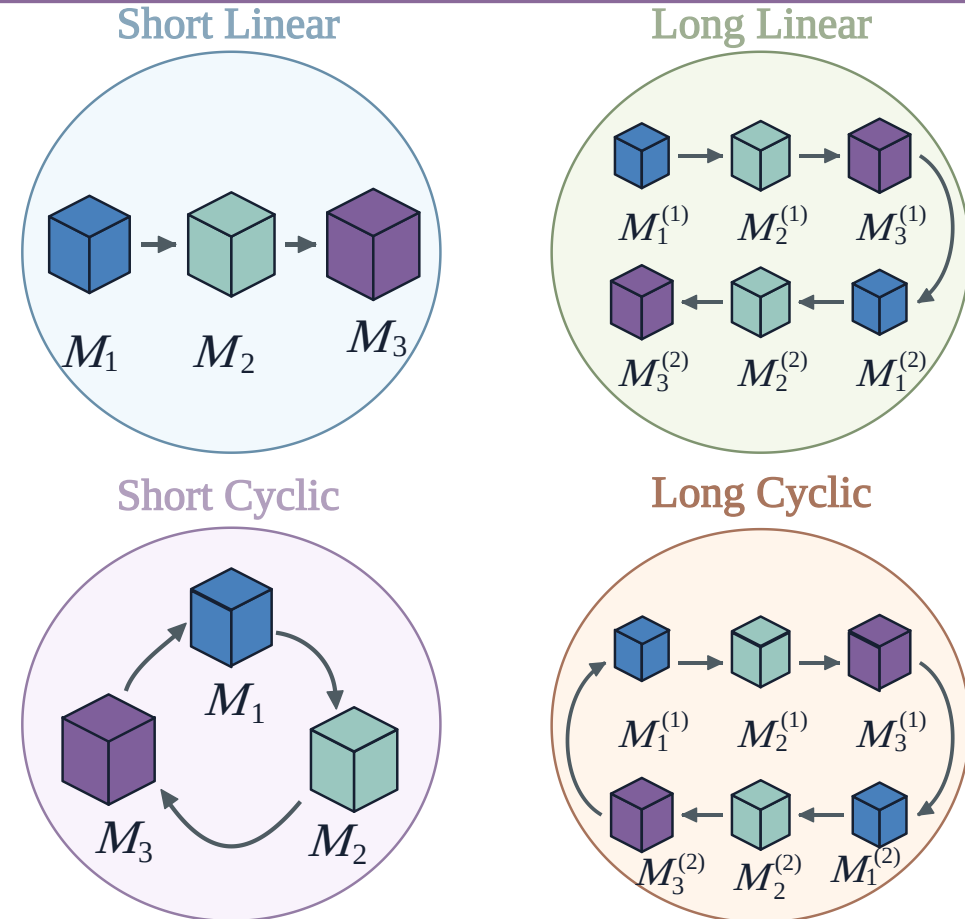


Model Slicing



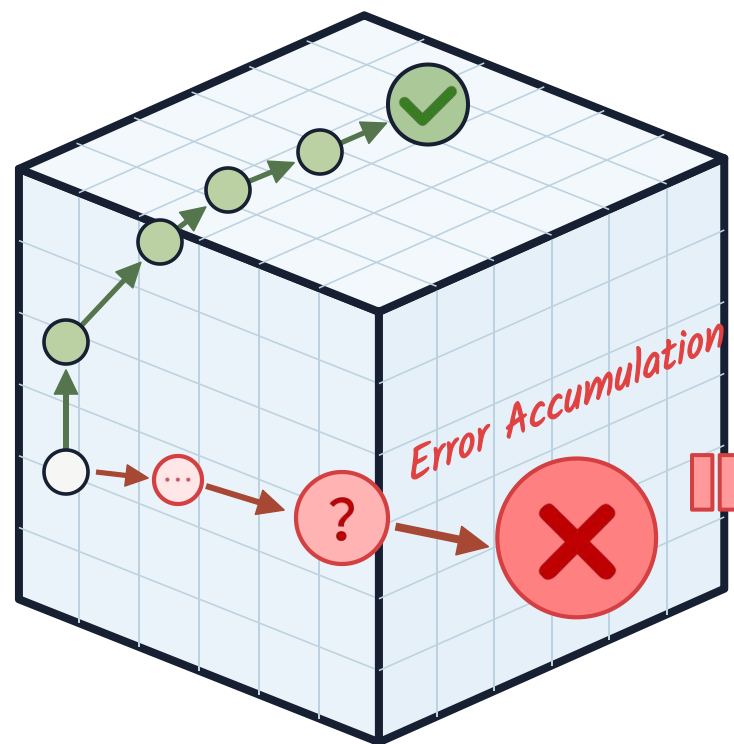
Pyramid Learning

Collaborative Reasoning



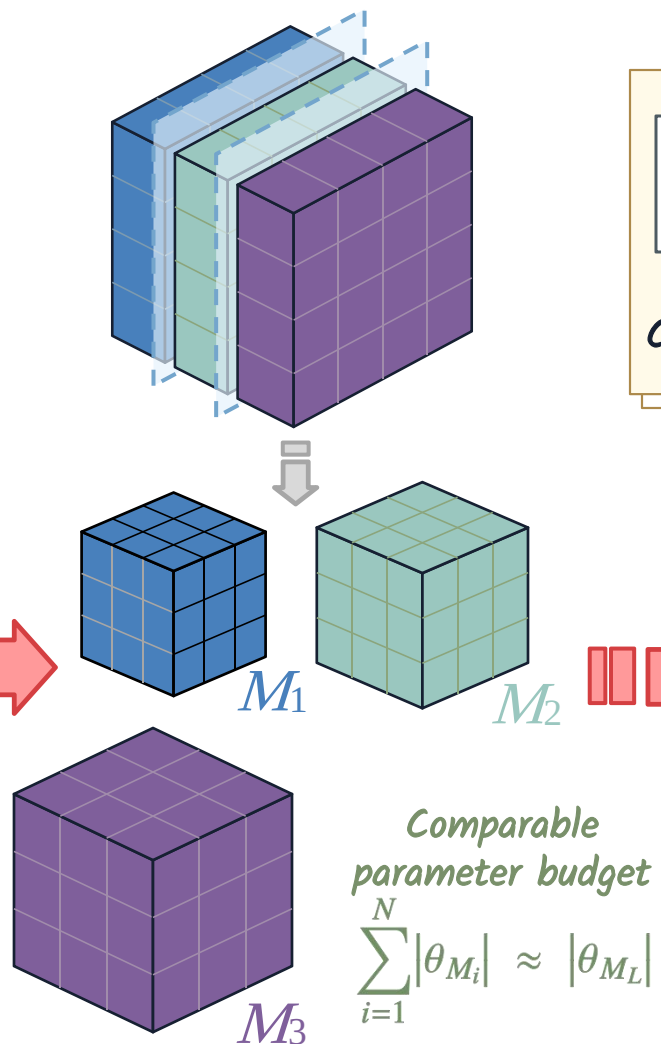
Emergent Collaboration Patterns

Single-Model Reasoning



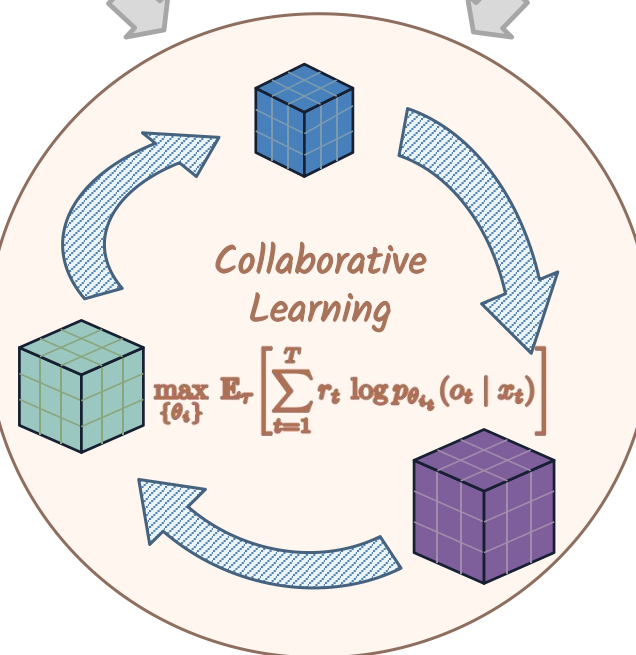
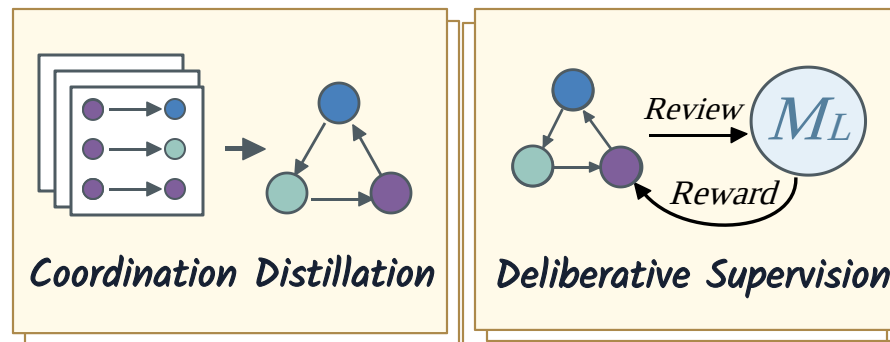
- Persistent reasoning blind spots
- Single-model iterative reasoning

Model Slicing



Reallocated
Parameter Budget

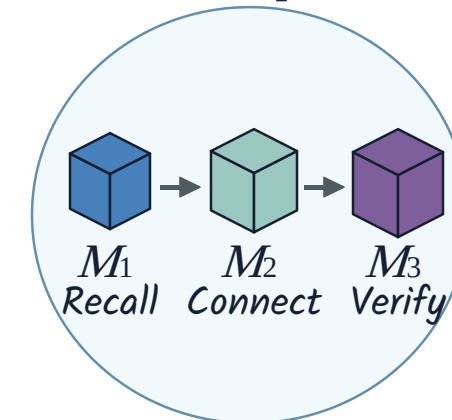
ModelCrew



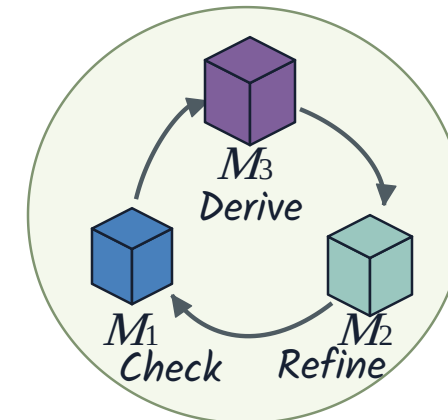
Learned Cross-Model Collaboration

Collaborative Reasoning

Multi-hop QA

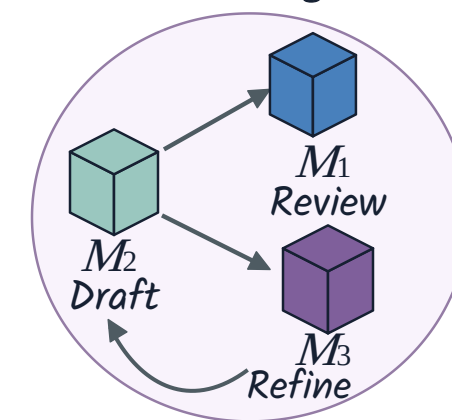


Math



Roles Emerge through Training

Coding



Open-ended Generation

